

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

**DDoS Attack Detection Using Network Traffic Features and
Machine Learning**

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**DDoS Attack Detection Using Network Traffic Features and Machine Learning**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**.

The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Distributed Denial of Service (DDoS) attacks are one of the major threats to modern network systems. These attacks overload network resources with malicious traffic, causing service disruption and affecting system availability. Traditional intrusion detection systems often fail to detect evolving and high-volume attacks effectively. This project presents a Machine Learning-based approach for detecting DDoS attacks using network traffic features. The dataset used contains 852,585 instances of normal and malicious traffic. Data preprocessing techniques such as removal of data leakage, label encoding, feature scaling, and correlation analysis were applied to improve model performance. Three Machine Learning algorithms — Logistic Regression, Random Forest, and XGBoost — were implemented and compared. Experimental results showed that ensemble models achieved very high accuracy, with XGBoost providing the best performance. The proposed system helps improve network security through accurate and automated DDoS attack detection.

1. INTRODUCTION

1.1 Problem Background

Distributed Denial of Service (DDoS) attacks are among the most dangerous cybersecurity threats in modern network systems. These attacks occur when attackers flood a server or network with a huge amount of fake traffic, making the service unavailable to legitimate users. DDoS attacks can affect websites, cloud platforms, banking systems, healthcare services, and government organizations, leading to financial losses and disruption of online services. With the rapid growth of cloud computing, IoT devices, and internet-based applications, cyber threats have increased significantly. Traditional intrusion detection systems mainly rely on predefined attack signatures and fixed rules. These systems are unable to detect new or evolving attacks efficiently, especially when attack patterns continuously change. ML algorithms can learn traffic behavior patterns from data and automatically distinguish between normal and malicious traffic. This helps improve detection accuracy and supports intelligent and automated network security systems. The increasing need for secure and reliable communication systems has created a demand for advanced DDoS detection methods. Therefore, this project focuses on using Machine Learning algorithms to analyze network traffic features and accurately detect DDoS attacks.

1.2 Objectives

The major objectives of this project are:

- To study and analyze DDoS attacks and their impact on network systems.
- To preprocess and clean the network traffic dataset for improved model performance.
- To remove unnecessary and leakage features from the dataset.
- To perform feature correlation analysis for identifying important traffic features.
- To implement Machine Learning techniques for DDoS attack detection.
- To train and compare Logistic Regression, Random Forest, and XGBoost models.
- To evaluate model performance using accuracy, precision, recall, and F1-score.
- To reduce false predictions and improve attack detection reliability.
- To generate visual analysis using graphs and confusion matrices.
- To develop an intelligent and scalable cybersecurity solution.

1.3 Proposed Solution

The proposed system is a Machine Learning-based DDoS attack detection framework that analyzes network traffic data and classifies traffic as normal or malicious. The system uses data preprocessing, feature engineering, correlation analysis, and Machine Learning algorithms to improve attack detection accuracy.

The workflow of the proposed system begins with dataset collection and preprocessing. During preprocessing, unnecessary features and data leakage are removed to improve the reliability of the models. Categorical features are converted into numerical values using label encoding, and feature scaling is applied for better model training.

After preprocessing, correlation analysis is performed to identify the most important network traffic features related to DDoS attacks. The processed dataset is then divided into training and testing sets using stratified sampling.

Three Machine Learning algorithms are implemented:

- Logistic Regression
- Random Forest
- XGBoost

The models are trained using network traffic data and evaluated using different performance metrics such as accuracy, precision, recall, and F1-score. Confusion matrices and graphs are also generated for better visualization and analysis of results.

The main goal of the proposed solution is to provide fast, accurate, and reliable DDoS attack detection for improving modern cybersecurity systems.

1.4 Key Features

The important features of the proposed system are:

- Automatic detection of DDoS attacks using Machine Learning algorithms.
- Analysis of network traffic data to identify suspicious activities.
- Removal of unnecessary and leakage features during preprocessing.
- Feature correlation analysis for better prediction performance.
- Comparison of multiple Machine Learning algorithms.
- High prediction accuracy with minimal false alarms.
- Visualization of results using graphs and confusion matrices.

- Improved network monitoring and cybersecurity support.
- Scalable framework suitable for future real-time applications.
- Reliable and efficient attack detection system.

1.5 Technology Stack Overview

The project is implemented using Python and various Machine Learning libraries. Python was selected because of its simplicity, flexibility, and strong support for data analysis and Machine Learning applications.

The following technologies and libraries were used in the project:

- Pandas and NumPy for data preprocessing and numerical analysis.
- Scikit-learn for implementing Logistic Regression and Random Forest models.
- XGBoost library for gradient boosting-based classification.
- Matplotlib and Seaborn for data visualization and graph generation.
- Jupyter Notebook and Visual Studio Code as the development environment.

These technologies helped in efficient data analysis, model training, visualization, and performance evaluation of the DDoS detection system.

2. METHODOLOGY

The methodology of the proposed system focuses on detecting DDoS attacks using Machine Learning techniques and network traffic analysis. The complete process includes dataset collection, data preprocessing, feature analysis, Machine Learning model training, and performance evaluation. The methodology was designed to improve attack detection accuracy and ensure reliable model performance.

2.1 Dataset Collection

A publicly available DDoS network traffic dataset was used in this project. The dataset contains both normal and malicious network traffic instances and is suitable for binary classification tasks. The dataset consists of approximately 852,585 instances with multiple traffic-related features extracted from packet headers and network flow statistics.

The dataset includes important features such as:

- Highest Layer
- Transport Layer
- Destination IP
- Source Port
- Destination Port
- Packet Length
- Packets/Time
- Target Variable

The target feature represents:

- 0 → Normal Traffic
- 1 → DDoS Attack Traffic

The dataset provides enough information for analyzing network traffic behavior and identifying attack patterns.

2.2 Data Preprocessing

Data preprocessing is one of the most important steps in Machine Learning because raw network data often contains unnecessary information, categorical values, and inconsistent data types. Proper preprocessing improves model performance and detection accuracy.

The following preprocessing steps were performed:

2.2.1 Removal of Data Leakage Features

Initial analysis showed that the “Source IP” feature had a very high correlation with the target variable. This could cause data leakage and unrealistic prediction results. Therefore, the Source IP feature was removed from the dataset to improve model generalization and reliability.

2.2.2 Label Encoding

Machine Learning models require numerical input data. Therefore, categorical features such as:

- Highest Layer
- Transport Layer
- Destination IP

were converted into numerical values using label encoding techniques.

2.2.3 Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

Stratified sampling was used to maintain equal class distribution in both training and testing datasets.

2.2.4 Feature Scaling

Feature scaling was performed to normalize feature values and improve model training performance. Standardization was applied to transform features into a common scale with zero mean and unit variance.

2.3 Correlation Analysis

Correlation analysis was performed to identify the most important features responsible for detecting DDoS attacks. Pearson correlation coefficient was used to measure the relationship between network traffic features and the target variable.

The analysis showed that:

- Packet Length
- Highest Layer
- Transport Layer
- Destination Port

were the most significant features for identifying attack traffic.

Correlation heatmaps were generated for visual analysis of feature relationships.

2.4 Machine Learning Models

Three Machine Learning algorithms were implemented and compared in this project. These models were selected because of their effectiveness in classification and cybersecurity applications.

2.4.1 Logistic Regression

Logistic Regression is a supervised Machine Learning algorithm used for binary classification problems. It predicts the probability of attack and normal traffic using a sigmoid function. The model is simple, fast, and effective for linear classification tasks.

2.4.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It handles high-dimensional data efficiently and reduces overfitting problems. Random Forest is highly effective for DDoS attack detection due to its robustness and feature importance capabilities.

2.4.3 XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced ensemble Machine Learning algorithm based on gradient boosting techniques. It provides high accuracy, faster processing, and better optimization compared to traditional models. XGBoost performed best among all implemented models in this project.

2.5 Performance Evaluation Metrics

The performance of Machine Learning models was evaluated using standard classification metrics derived from the confusion matrix.

2.5.1 Accuracy

Accuracy measures the overall correctness of the model and represents the percentage of correctly classified instances.

2.5.2 Precision

Precision measures how many predicted attack instances were actually correct. It helps evaluate the reliability of positive predictions.

2.5.3 Recall

Recall measures the ability of the model to correctly identify actual attack traffic. It is important for minimizing missed attacks.

2.5.4 F1-Score

F1-Score is the harmonic mean of precision and recall. It provides a balanced measure of model performance.

2.5.5 Confusion Matrix

Confusion matrices were used to visualize prediction performance and analyze false positives and false negatives generated by each model. The evaluation metrics helped compare the performance of Logistic Regression, Random Forest, and XGBoost models effectively.

3. TOOLS AND TECHNOLOGIES

Different tools, libraries, and technologies were used in this project for data preprocessing, Machine Learning model implementation, visualization, and performance evaluation. These technologies helped in developing an efficient and reliable DDoS attack detection system.

Category	Technology Used
Programming Language	Phyton
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost
Visualization Tools	Matplotlib, Seaborn
Development Environment	Jupyter Notebook, VS Code

4. IMPLEMENTATION

The implementation phase focuses on developing the DDoS attack detection system using Machine Learning techniques and network traffic analysis. The complete implementation process includes dataset loading, preprocessing, feature analysis, model training, testing, and visualization of results. The system was implemented using Python and various Machine Learning libraries in a structured manner to ensure accurate and reliable attack detection.

4.1 Dataset Loading and Processing

The first step of implementation involved loading the DDoS network traffic dataset into the system using the Pandas library. The dataset contained approximately 852,585 instances consisting of both normal and malicious network traffic.

After loading the dataset, several preprocessing steps were performed:

- Removal of unnecessary features
- Handling categorical attributes
- Label encoding
- Feature scaling
- Data cleaning and formatting

The “Source IP” feature was removed because it caused data leakage and could negatively affect model generalization. The processed dataset was then prepared for Machine Learning model training and testing.

4.2 Feature Extraction and Analysis

Feature extraction and analysis were performed to identify the most important network traffic characteristics responsible for DDoS attack detection. Correlation analysis was applied using Pearson correlation coefficients to measure the relationship between features and the target variable.

The analysis identified:

- Packet Length
- Highest Layer
- Transport Layer
- Destination Port

as the most important features for distinguishing between normal and malicious traffic. Correlation heatmaps and graphs were generated for visual representation of feature relationships.

4.3 Machine Learning Model Training

Three Machine Learning models were implemented and trained using the processed dataset.

Logistic Regression

Logistic Regression was implemented as a baseline classification model for detecting DDoS attacks. The model predicts the probability of attack traffic using a sigmoid function and performs binary classification.

Random Forest

Random Forest was implemented using multiple decision trees for improving classification accuracy and reducing overfitting. The model combines predictions from several trees to generate the final output.

XGBoost

XGBoost was implemented as an advanced ensemble learning algorithm based on gradient boosting techniques. It improves prediction performance by sequentially correcting errors generated by previous models.

The dataset was divided into:

- 80% training data
- 20% testing data

Stratified train-test splitting was used to maintain balanced class distribution during model training.

4.4 Model Testing and Evaluation

After training, all Machine Learning models were tested using the testing dataset. The performance of each model was evaluated using standard classification metrics.

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score

Confusion matrices were generated for all models to analyze:

- True Positives
- True Negatives
- False Positives
- False Negatives

The evaluation results showed that ensemble models such as Random Forest and XGBoost achieved significantly better performance compared to Logistic Regression.

Table 1:Model Accuracy Results

Model	Accuracy
Logistic Regression	99.34%
Random Forest	99.99%
XGBoost	99.995%

XGBoost achieved the highest accuracy and provided the most reliable DDoS attack detection performance.

4.5 Graph and Visualization Generation

Visualization techniques were used to analyze model performance and feature relationships more effectively.

The following visualizations were generated:

- Correlation Heatmap
- Model Accuracy Comparison Graph
- Confusion Matrices
- Feature Mean Comparison Graphs

Matplotlib and Seaborn libraries were used for generating these visualizations.

These graphs helped in:

- Understanding feature importance
- Comparing Machine Learning model performance
- Identifying prediction patterns
- Improving result interpretation

The implementation process successfully developed a reliable and efficient Machine Learning-based DDoS attack detection system capable of identifying malicious network traffic with high accuracy.

Fig. 1. Correlation Heatmap of Features

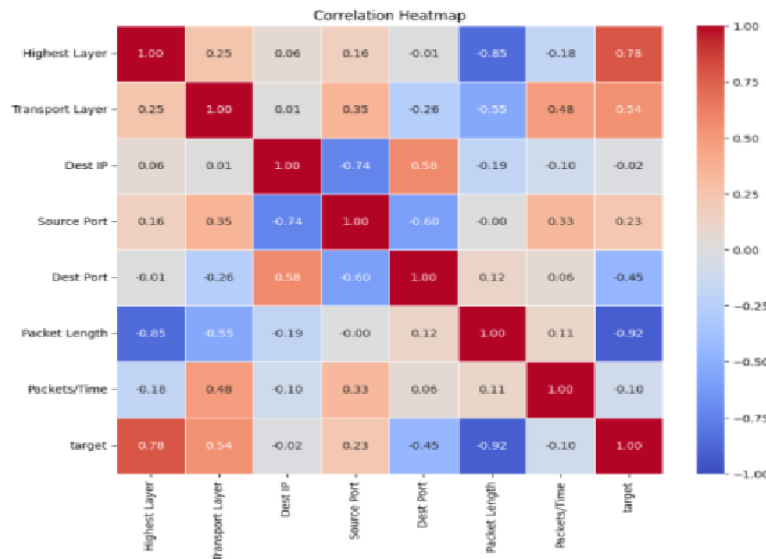


Fig. 2. Model Accuracy Comparison

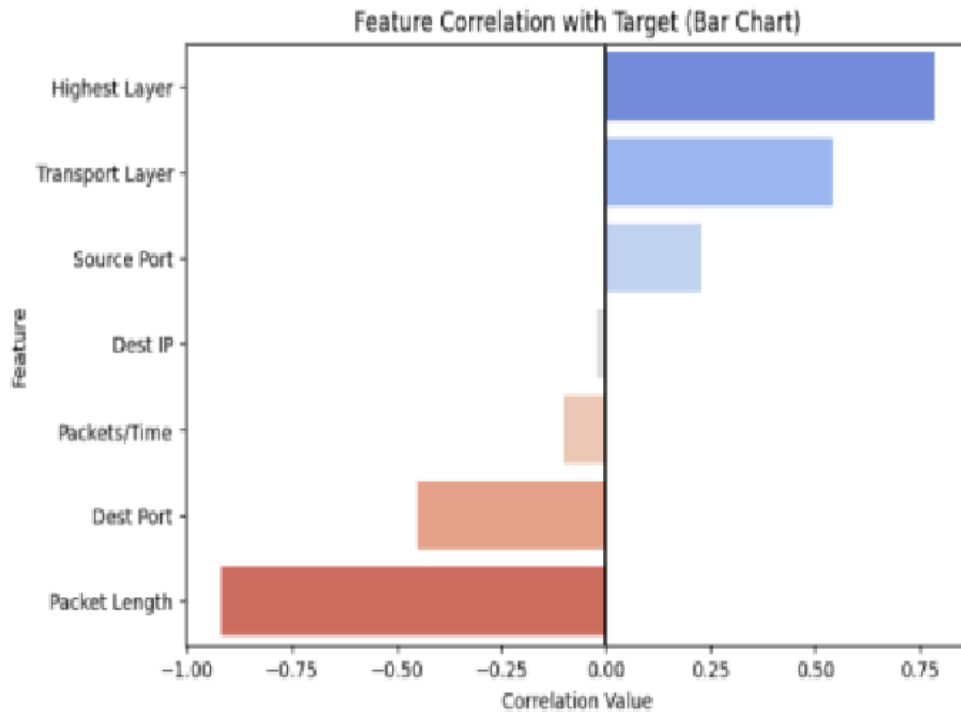


Fig 3. Logistic Regression Confusion Matrix

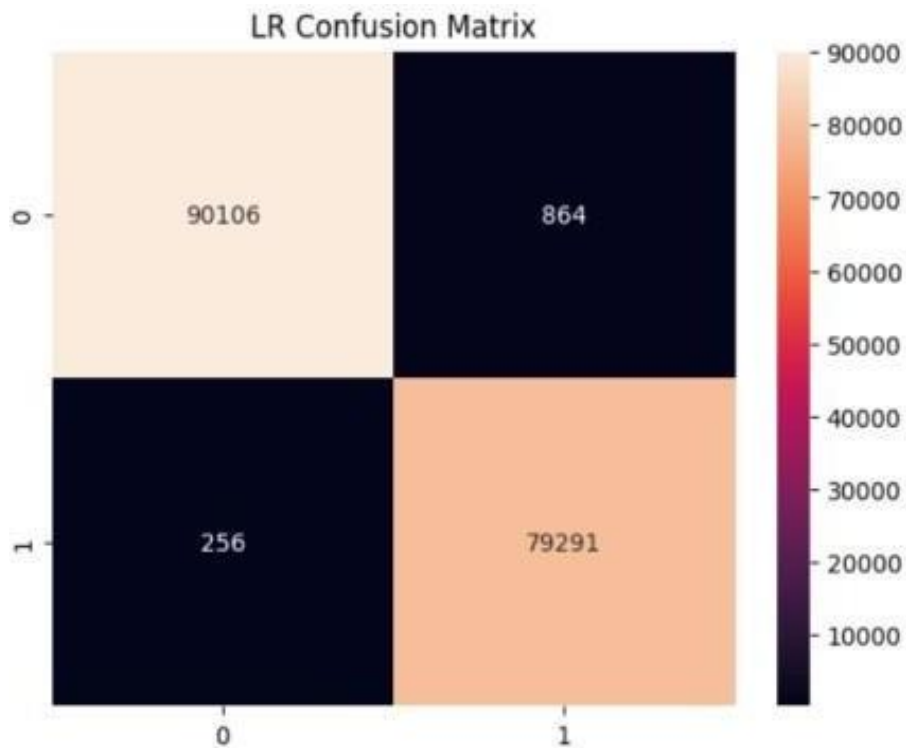


Fig. 4. Random Forest Confusion Matrix

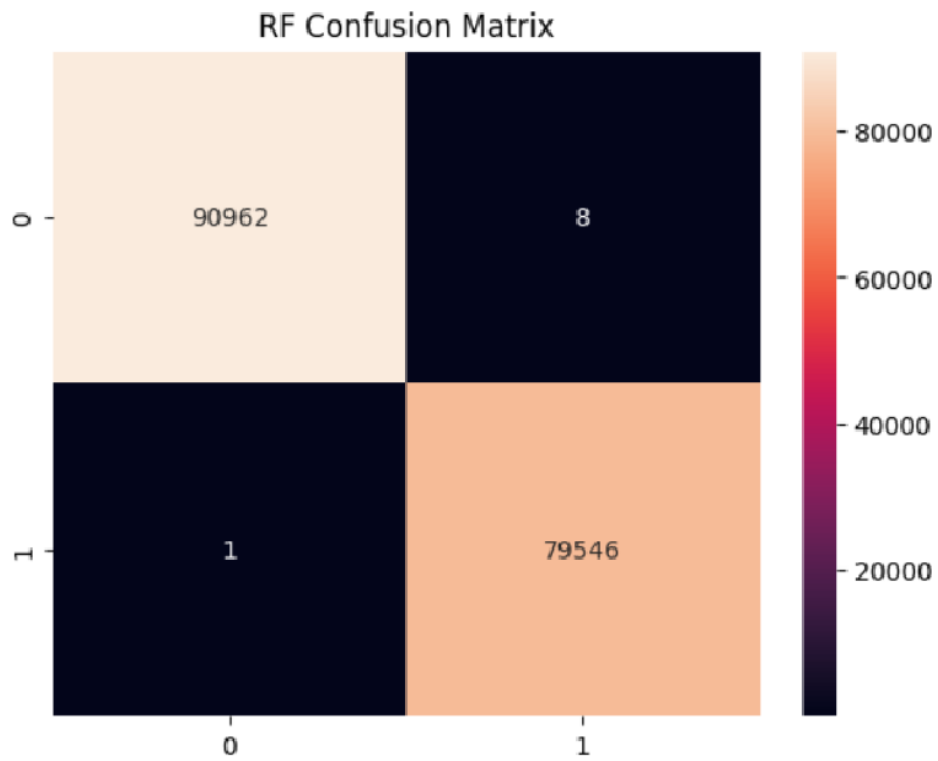
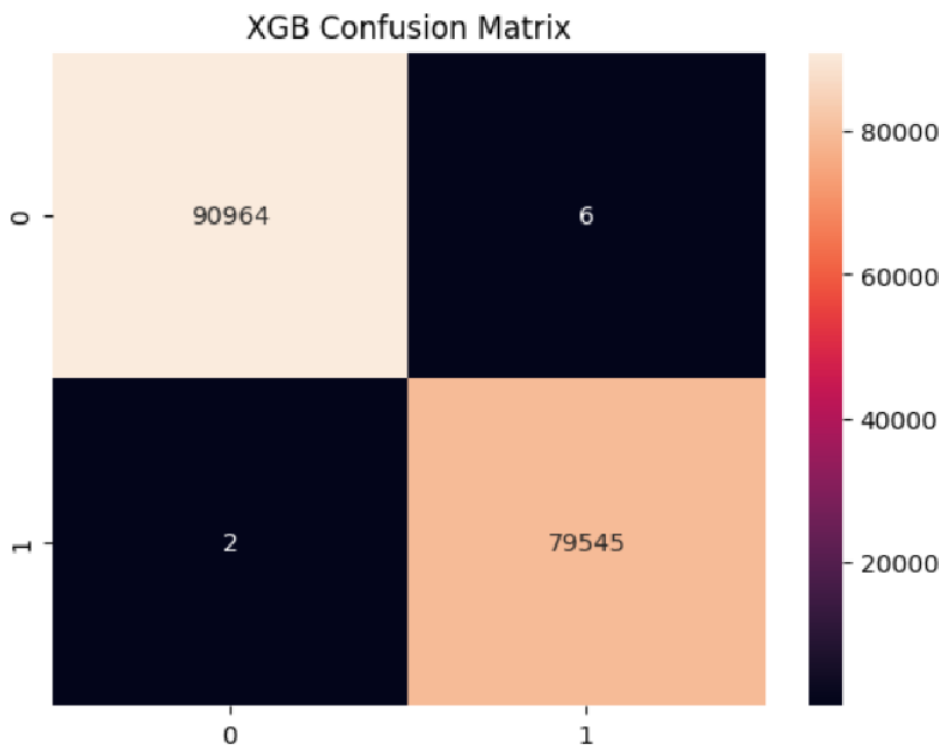


Fig. 5. XGBOOST Confusion Matrix



5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

This section presents the major findings, performance analysis, and experimental results obtained from the implementation of Machine Learning models for DDoS attack detection. The results demonstrate the effectiveness of preprocessing techniques, feature analysis, and ensemble Machine Learning algorithms in accurately identifying malicious network traffic. The models were evaluated using different performance metrics such as accuracy, precision, recall, and F1-score. Various graphs and confusion matrices were also generated for detailed analysis and visualization of result

5.1 Correlation Analysis Results

Correlation analysis was performed to identify the most important features responsible for detecting DDoS attacks. Pearson correlation coefficient was used to measure the relationship between network traffic features and the target variable.

The analysis revealed that the following features had the highest impact on attack detection:

Feature	Correlation with Target
Packet Length	-0.917677
Highest Layer	0.782798
Transport Layer	0.543415
Destination Port	-0.449707
Source Port	0.227624
Packets/Time	-0.100678
Destination IP	-0.023216

Key Observations

- Packet Length showed the strongest correlation with attack traffic.
- Highest Layer and Transport Layer were highly useful in identifying attack patterns.
- Destination Port also played an important role in differentiating attack and normal traffic.
- Some features showed weaker correlations but still contributed to overall model performance.

A correlation heatmap was generated to visualize the relationships between all features and the target variable.

5.2 Model Performance Results

Three Machine Learning algorithms were trained and evaluated:

- Logistic Regression
- Random Forest
- XGBoost

The models were compared using accuracy, precision, recall, and F1-score metrics.

Table 2: Performance Comparison of ML Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.993432	0.99	0.99	0.99
Random Forest	0.999947	1.00	1.00	1.00
XGBoost	0.999953	1.00	1.00	1.00

Performance Analysis

- Logistic Regression achieved very high accuracy and performed well for binary classification.
- Random Forest significantly improved detection performance due to ensemble learning techniques.
- XGBoost achieved the highest accuracy among all implemented models.
- Ensemble models provided better reliability and lower false predictions compared to linear models.

The results clearly indicate that ensemble learning techniques are highly effective for DDoS attack detection.

5.3 Confusion Matrix Analysis

Confusion matrices were generated for all Machine Learning models to analyze classification performance in detail.

The confusion matrices helped in evaluating:

- True Positives (Correctly detected attacks)
- True Negatives (Correctly identified normal traffic)
- False Positives
- False Negatives

Observations

- Logistic Regression produced a small number of false predictions.
- Random Forest and XGBoost achieved almost perfect classification performance.
- False positive and false negative rates were extremely low in ensemble models.
- XGBoost provided the most balanced and reliable predictions.

The confusion matrix visualizations demonstrated the efficiency of the proposed system in distinguishing attack traffic from normal traffic.

5.4 Feature Mean Analysis

Feature mean comparison was performed to study behavioral differences between normal and malicious traffic.

The analysis showed that:

- Attack traffic had significantly different packet length patterns.
- Transport Layer and protocol-related features varied greatly between attack and normal traffic.
- Port usage behavior also differed noticeably during DDoS attacks.

These observations helped validate the importance of selected features and supported the effectiveness of feature engineering techniques used in the project.

5.5 Key Findings

The major findings obtained from this project are:

- Machine Learning techniques can accurately detect DDoS attacks using network traffic features.
- Data preprocessing and feature selection significantly improve model performance.
- Ensemble models such as Random Forest and XGBoost outperform traditional linear models.
- Packet Length was identified as the most important feature for attack detection.
- XGBoost achieved the highest prediction accuracy among all implemented models.
- Confusion matrix analysis showed extremely low false prediction rates.
- The proposed system provides reliable and scalable cybersecurity support for modern network environments.

The overall results demonstrate that Machine Learning-based DDoS detection systems can effectively improve network security and support intelligent cyberattack prevention mechanisms.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

In this project, a Machine Learning-based approach for detecting DDoS attacks using network traffic features was successfully implemented and evaluated. The system analyzed network traffic data and classified traffic as normal or malicious using various Machine Learning algorithms. The dataset used in this project contained approximately 852,585 instances of network traffic data. Several preprocessing techniques such as removal of data leakage features, label encoding, feature scaling, and correlation analysis were applied to improve model performance and reliability.

Three Machine Learning models — Logistic Regression, Random Forest, and XGBoost — were trained and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. Experimental results showed that ensemble learning models achieved significantly better performance compared to traditional linear models.

Among all implemented models:

- Logistic Regression achieved an accuracy of 99.34%
- Random Forest achieved an accuracy of 99.99%
- XGBoost achieved the highest accuracy of 99.995%

The results demonstrated that Machine Learning techniques can effectively identify DDoS attacks with very high accuracy and minimal false predictions. Correlation analysis also identified important traffic features such as Packet Length, Highest Layer, and Destination Port, which played a major role in attack detection.

Overall, the proposed system provides an intelligent, reliable, and scalable solution for improving network security and DDoS attack detection in modern network environments.

6.2 Future Scope

Although the proposed system achieved excellent results, several improvements and future enhancements can be made to further increase its effectiveness and practical usability.

Future improvements may include:

- Development of real-time DDoS attack detection systems.
- Implementation of multi-class classification for detecting different attack types.
- Integration of deep learning techniques for advanced threat detection.

- Enhancement of security for IoT and cloud-based environments.
- Handling larger and real-world network traffic datasets.
- Development of lightweight models for edge and IoT devices.
- Integration of automated alert and notification systems.
- Creation of live monitoring dashboards for network administrators.
- Improvement in detection speed and scalability.
- Application of transfer learning techniques for better adaptability to new attack patterns.

These future enhancements can help develop more intelligent, faster, and highly secure cybersecurity systems capable of handling evolving DDoS attacks in real-world environments.

7. REFERENCES

- i. A. Sharma and H. Babbar, "BoT-IoT: Detection of DDoS Attacks in Internet of Things for Smart Cities," in *2023 10th International Conference on Computing for Sustainable Global Development (INDIACom)*, New Delhi, India, 2023, pp. 1-6.
- iii. F. B. Saghezchi, G. Mantas, M. A. Violas, A. M. de Oliveira Duarte, and J. Rodriguez, "Machine Learning for DDoS Attack Detection in Industry 4.0 CPPSs," *Electronics*, vol. 11, no. 4, p. 602, Feb. 2022.
- iv. J. Pei, Y. Chen, and W. Ji, "A DDoS Attack Detection Method Based on Machine Learning," *Journal of Physics: Conference Series*, vol. 1237, no. 3, p. 032040, 2019.
- vii. A. Dremov and A. Volokyta, "DDoS Attack Detection with Data Imperfections Using Machine Learning Algorithms," *Eastern-European Journal of Enterprise Technologies*, vol. 6, no. 9, pp. 1-11, 2024.
- viii. T. E. Ali, Y. W. Chong, and S. Manickam, "Machine Learning Techniques to Detect a DDoS Attack in SDN: A Systematic Review," *Applied Sciences*, vol. 13, no. 5, p. 3183, Mar. 2023.
- ix. T. Saito, Q. Wu, and A. Kanai, "Improving DDoS Attack Detection via DNN and SHAP-based Feature Selection in Transfer Learning," in *2025 IEEE 14th Global Conference on Consumer Electronics (GCCE)*, Osaka, Japan, 2025, pp. 1-4.

8. APPENDICES

8.1 Appendix A – Dataset Information

- Dataset contains 852,585 instances of normal and attack traffic.
- Features include Packet Length, Destination Port, Transport Layer, etc.

8.2 Appendix B – Machine Learning Models Used

- Logistic Regression
- Random Forest
- XGBoost

8.3 Appendix C – Graphs and Confusion Matrices

- Correlation Heatmap
- Accuracy Comparison Graph
- Confusion Matrices

8.4 Appendix D – Software and Libraries

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- Seaborn